(202111.2) Edgecore SONiC Quick Start Guide

This quick start guide provides an end-to-end setup process for installing and running SONiC on an Edgecore-supported switch, as well as a collection of example commands for getting started after installation is complete.

Prerequisites

This guide assumes you have an intermediate level of Linux/Unix knowledge. You need to be familiar with basic Linux/Unix commands, Linux/Unix file permissions, and process monitoring.

You must have access to a Linux or UNIX shell. If you are running Windows, connect to the Linux environment using the Putty SSH client to access the Edgecore device, and use the cmd prompt to connect to the console port of the device for interacting with SONiC.

Quick Start

The document provides some insights into the following: (You may skip this if you are an experienced user.)

- (202111.2) Install, Upgrade, and Remove Edgecore SONiC
- (202111.2) Setting Username and Password
- (202111.2) Configuring IPv4/IPv6 Addresses for Ports
- (202111) Enabling the Zero Touch Provisioning (ZTP) Service
- (202111.2) Backing Up and Restoring the Default Configuration, and (202111.2) Restoring the Configuration to Factory Defaults
- (202111.2) Contacting Technical Support

(202111.2) Install, Upgrade, and Remove Edgecore SONiC

This section describes step-by-step procedures for installing, upgrading, and removing Enterprise SONiC Distribution by Edgecore Networks (Edgecore SONiC) on a switch, and managing the installed software images.

Edgecore SONiC Installation

Once an Edgecore SONiC image is available, it can be installed using either of the two following methods:

- Edgecore SONiC Installation via TFTP/HTTP
- Edgecore SONiC Installation via USB

Edgecore SONiC Installation via TFTP/HTTP

This sub-section explains how to transfer an Edgecore SONiC image from a remote server to a device and install it. Make sure to connect the TFTP/HTTP server to the management port of the switch before starting the installation procedure.

Fig 1. Connecting a TFTP/HTTP server to a switch

Step 1. Enter the ONIE install mode.

> Note
>
> The switch is preprogrammed to enter ONIE installation mode if there is no current installed NOS.

Fig 2. ONIE sub menus via TFTP/HTTP

ONIE auto-discovery is enabled by default. Use the **onie-discovery-stop** command to stop the ONIE Service Discovery.

Example

```
ONIE:/ # onie-discovery-stop
discover: installer mode detected.
Stopping: discover... done.
```

Note

onie-discovery-stop is an optional command. It does not affect the installation whether it is used or not.

Step 2. Set the IP address for the switch management port.

For each of the remote installations to work properly, an IP address is needed on the switch. Use the **ifconfig** command to set the IP address.

Example

```
ONIE:/ # ifconfig eth0 192.168.1.2 netmask 255.255.255.0
```

Step 3. Install the Edgecore SONiC image from a remote URL via HTTP or TFTP.

Here, we assume you have uploaded the SONiC image onto an HTTP/TFTP server (example: 192.168.1.3). Use the **onie-nos-install** command to install the Edgecore SONiC software image.

Example

```
ONIE:/ # onie-nos-install tftp://192.168.1.3/Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178.
bin
```

If the installation is successful, the device will reboot automatically and boot up with SONiC.

After completing the installation, it is recommended to change the password to avoid mishandling of the credentials. See the procedure in (202111.2) Setting Username and Password for more details.

Edgecore SONiC Installation via USB

This sub-section explains the procedure to install an Edgecore SONiC image from a USB drive.

Caution

Only software versions **SONiC.Edgecore-SONiC_20200722_070543_ec201911_141** and later supports USB installation. It is recommended to use the TFTP/HTTP procedure for older software versions.

Step 1. Copy the installer file to a USB flash drive.

Make sure the Edgecore SONiC installer file has the name **onie-installer-x86_64**.

Step 2. Reboot the switch and plug the USB drive into the switch.

Step 3. Enter ONIE install mode.

Fig 3. ONIE sub menus via USB drive

Step 4. Wait for ONIE to discover the USB and start automatically uploading the new software from the USB drive.

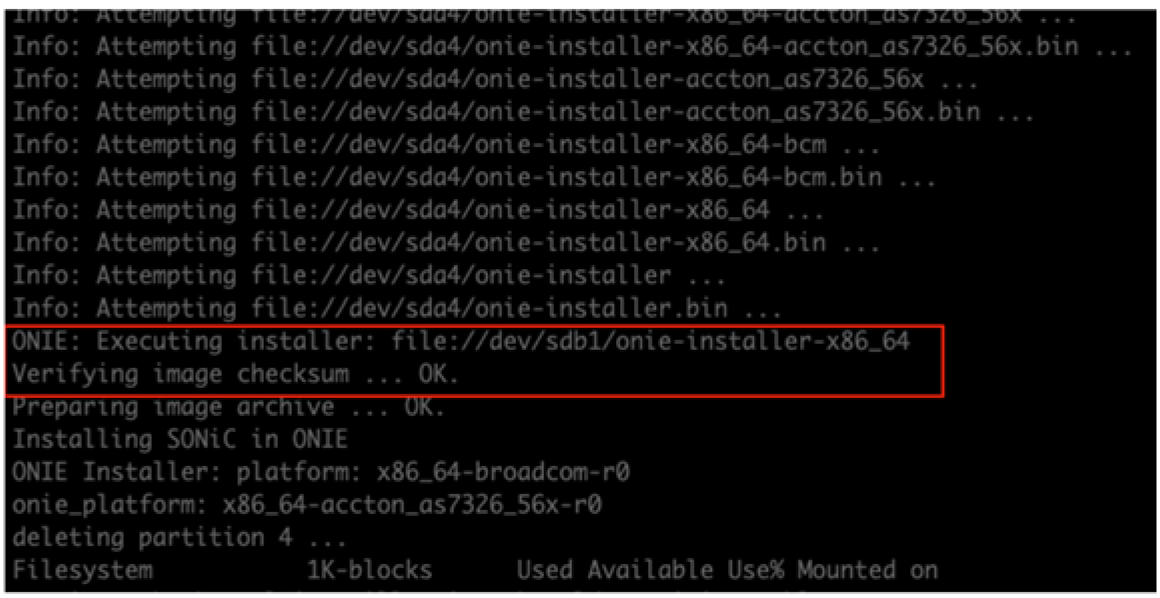


Fig 4. Installing SONiC using USB

If the installation is successful, the device will reboot automatically and boot up with SONiC.

After completing the installation, it is recommended to change the password to avoid mishandling of the credentials. See the procedure in (202111.2) Setting Username and Password for more details.

Edgecore SONiC Upgrade

Edgecore SONiC software can be installed using two methods; using the "ONIE Installer" or using the "sonic-installer tool." "ONIE Installer" shall be used as shown in the Edgecore SONiC Installation section, and "sonic-installer" shall be used as given below. Follow this procedure to install a new Edgecore SONiC image in the alternate image partition.

Step 1. Check the existing installed software images.

Use the **sonic-installer list** command to check the currently installed sonic software images. It displays a list of installed images, currently running images, and the image set to be booted up on the next reboot.

Example

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
Next: SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
Available:
SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
```

Step 2. Set the management port IP address.

(For more information, see (202111.2) Configuring IPv4/IPv6 Addresses for Ports.)

Example

```
admin@sonic:~$ sudo ip address add 192.168.1.2/24 dev eth0
```

Step 3. Upgrade the software image from a remote HTTP server.

Use the **sonic-installer install** command to install a new image in the alternate image partition. This command takes a path to an installable SONiC image or URL and installs the image.

Example

```
admin@sonic:~$ sudo sonic-installer install http://192.168.1.3/SONiC-OS-Edgecore-
SONiC_20200827_110345_ec201911_2020-aug_enhanced_178.bin -y
```

> Caution
>
> You can only use HTTP to upgrade the Edgecore SONiC software.

Step 4. Check the installed software images.

The newly installed image is set to "Next", which will boot up on the next reboot.

Example

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
Next: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Available:
SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
```

Step 5. Reboot the switch.

Example

```
admin@sonic:~$ sudo reboot
```

Step 6. Check the status of the installed software images.

Example

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Next: SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
Available:
SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
```

Changing the Default Boot-Up Image

This subsection describes how to modify the image that will boot up by default in all subsequent reboots. Follow the below procedure to set the default boot-up software image.

Step 1. Check the status of the installed software images.

Example

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Next: SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
Available:
SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
```

Step 2. Change the default boot-up image.

Use the **sonic-installer set-default** command to change the default boot-up software image.

Example

```
admin@sonic:~$ sudo sonic-installer set-default SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-
aug_enhanced_178
Command: grub-set-default --boot-directory=/host 1
```

Step 3. Check the status of the installed software images.

In the installed software image list, the "Next" boot-up image is changed to a new image that was set using the **sonic-installer set-default** command in the previous step.

Example

```
admin@sonic:~$ sudo sonic_installer list
Current: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Next: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Available:
SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
```

Note

You can also choose the boot-up image directly from the GRUB menu, but this does not change the default boot-up image.

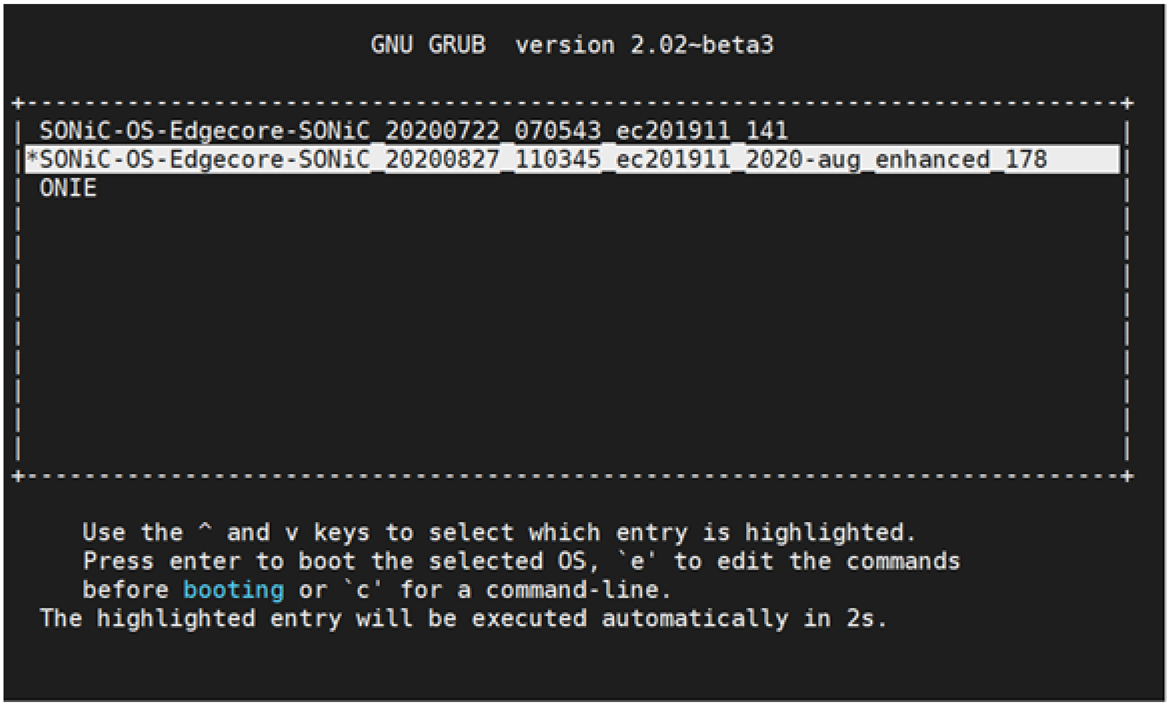


Fig 5. Boot-up image set from GRUB

Removing an Installed SONiC Image

To remove a SONiC image from the storage, follow the steps below.

Step 1. Check the status of the installed software images.

Example

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Next: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Available:
SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141
```

Step 2. Remove the software image.

Use the **sonic-installer remove** command to remove a SONiC image from the storage.

Example

```
admin@sonic:~$ sudo sonic-installer remove SONiC-OS-Edgecore-SONiC_20200722_070543_ec201911_141 –y
```

Step 3. Check the status of the installed software images.

Example

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Next: SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
Available:
SONiC-OS-Edgecore-SONiC_20200827_110345_ec201911_2020-aug_enhanced_178
```

Note

The system will not allow you to remove the currently running image.

(202111.2) Setting Username and Password

This section describes how to configure passwords for Edgecore SONiC and provides some insights into the following:

- Default Username and Password
- Changing the Password
- Changing the SUDO Password

Default Username and Password

When you first log in to Edgecore SONiC, the default username and password are as follows:

- **Username:** admin
- **Password:** YourPaSsWoRd

Changing the Password

Step 1. Use the following command to see the available options for configuring a password.

Example

```
admin@sonic:~$ passwd -h
Usage: passwd [options] [LOGIN]
Options:
-a, --all report password status on all accounts
-d, --delete delete the password for the named account
-e, --expire force expire the password for the named account
-h, --help display this help message and exit
-k, --keep-tokens change password only if expired
-i, --inactive INACTIVE set password inactive after expiration to INACTIVE
-l, --lock lock the password of the named account
-n, --mindays MIN_DAYS set minimum number of days before password change to MIN_DAYS
-q, --quiet quiet mode
-r, --repository REPOSITORY change password in REPOSITORY repository
-R, --root CHROOT_DIR directory to chroot into
-S, --status report password status on the named account
-u, --unlock unlock the password of the named account
-w, --warndays WARN_DAYS set expiration warning days to WARN_DAYS
-x, --maxdays MAX_DAYS set maximum number of days before password change to MAX_DAYS
admin@sonic:~$
```

Step 2. Use the Linux command passwd to change the login password. The example below shows how to change the password for the user name “admin".

Example

```
admin@sonic:~$ sudo passwd admin
Enter new UNIX password: xxxx << enter new password here
Retype new UNIX password: xxxx << enter new password here
passwd: password updated successfully
```

For more details regarding password limitations and restrictions, refer to (202111) Password limitation or restriction in SONiC => give a new title for this page

Changing the SUDO Password

By default, when you access commands through the sudo group, the system does not request a password.

Example

```
admin@sonic:~$ sudo sed -n "50p" /etc/sudoers
%sudo ALL=(ALL:ALL) NOPASSWD: ALL
```

If you want to request a password every time you run a sudo group command, you can modify it as follows.

Example

```
admin@sonic:~$ sudo visudo
%sudo ALL=(ALL:ALL) ALL
```

(202111.2) Configuring IPv4/IPv6 Addresses for Ports

This section describes step-by-step procedures for configuring IPv4 and IPv6 addresses for the management port and front-panel ports.

> **Note**
>
> All the configuration commands need root privileges to execute, and the commands are case-sensitive. Show commands can be executed by all users without root privileges. Root privileges can be obtained either by using the "sudo" keyword in front of all config commands, or by going to the root prompt using "sudo -i".

Configuring a Management Port (eth0) IP Address

A DHCP server automatically allocates and assigns an IP address to the management port. However, you can configure an IP address for the management port by following the steps below.

Step 1. Configure an IP address on management port "eth0".

Use **config interface ip add** command to configure an IPv4 or IPv6 address.

Example

```
admin@sonic:~$ sudo config interface ip add eth0 188.188.97.16/16 188.188.1.1
admin@sonic:~$ sudo config interface ip add eth0 2001::8/64 2001::1
```

Step 2. Check the configured IP addresses.

Use the **show ip interfaces** and **show ipv6 interfaces** commands to display the configured IPv4 addresses and IPv6 addresses, as shown in the examples below.

Also, the **show management interface address** command can be used to display the IPv4 and IPv6 management addresses, as shown in the example below.

Example

```
admin@sonic:~$ show ip interfaces
Interface    Master    IPv4 address/mask    Admin/Oper    BGP Neighbor    Neighbor
-----------  --------  -------------------  ------------  --------------  ---------
Loopback0              10.1.0.1/32          up/up         N/A             N/A
docker0                240.127.1.1/24       up/down       N/A             N/A
eth0                   188.188.97.16/16     up/up         N/A             N/A
lo                     127.0.0.1/8          up/up         N/A             N/A

admin@sonic:~$ show ipv6 interfaces
Interface    Master    IPv6 address/mask                       Admin/Oper    BGP Neighbor    Neighbor IP
-----------  --------  --------------------------------------  ------------  --------------  -------------
Bridge                 fe80::4020:6aff:fee4:b0a3%Bridge/64     up/down       N/A             N/A
Ethernet0              fe80::ba6a:97ff:fee2:479c%Ethernet0/64  up/up         N/A             N/A
Ethernet1              fe80::ba6a:97ff:fee2:479c%Ethernet1/64  up/up         N/A             N/A
Ethernet48             fe80::ba6a:97ff:fee2:479c%Ethernet48/64 up/up         N/A             N/A
Ethernet52             fe80::ba6a:97ff:fee2:479c%Ethernet52/64 up/up         N/A             N/A
Loopback0              fe80::243e:e6ff:fee7:5ca1%Loopback0/64  up/up         N/A             N/A
eth0                   2001::8/64                              up/up         N/A             N/A
                       fe80::ba6a:97ff:fee2:479c%eth0/64
lo                     ::1/128                                 up/up         N/A             N/A

admin@sonic:~$ show management_interface address
Management IP address = 188.188.97.16/16
Management Network Default Gateway = 188.188.1.1
Management IP address = 2001::8/64
Management Network Default Gateway = 2001::1
```

Note

The **show management_interface address** command and the **BGP Neighbor information** command are not supported in the ecSONiC-201904 version.

Step 3. Save the settings to "config_db.json".

Use the **config save -y** command to save the configured IP addresses to the "**config_db.json**" file.

Example

```
admin@sonic:~$ sudo config save -y
```

Note

The modified **config_db.json** content will not be saved into the /etc/sonic/config_db.json file automatically. In order to do that, a **config save** command needs to be manually executed as shown in step 3.

Step 4. Check IP address settings in "config_db.json".

Use the **vi /etc/sonic/config_db.json** command to check the IP address settings in the "**config_db.json**" file.

Example

```
admin@sonic:$ sudo vi /etc/sonic/config_db.json
{
   ...omitted
   "MGMT_INTERFACE": {
       "eth0|188.188.97.16/16": {
           "gwaddr": "188.188.1.1"
       },
       "eth0|2001::8/64": {
           "gwaddr": "2001::1"
       }
     },
     ...omitted
}
```

Configuring a Front-Panel Port IP Address

Unlike a management port IP address, a front-panel port IP address should be set manually. You can configure an IP address for a front-panel port by following the steps below.

Step 1. Configure an IP address on a front-panel port.

Use the **config interface ip add** command to configure an IPv4 or IPv6 address on a front-panel port. The example below shows IPv4/IPv6 addresses configured for interface "Ethernet0".

Example

```
admin@sonic:~$ sudo config interface ip add Ethernet0 192.168.1.1/24
admin@sonic:~$ sudo config interface ip add Ethernet0 2002::1/64
```

> **Caution**
>
> Make sure to specify the correct interface name, otherwise the command does not work. There are no warning messages for an incorrect command.

Step 2. Check the Configured IP addresses.

Use the **show ip interfaces** and **show ipv6 interfaces** commands to display the configured IPv4 addresses and IPv6 addresses, as shown in the examples below.

Example

```
admin@sonic:~$ show ip interfaces
Interface   IPv4 address/mask   Admin/Oper
----------- ------------------- ------------
Ethernet0   192.168.1.1/24      up/down
docker0     240.127.1.1/24      up/down
eth0        188.188.97.32/16    up/up
lo          127.0.0.1/8         up/up
            10.1.0.1/32

admin@sonic:~$ show ipv6 interfaces
Interface   IPv6 address/mask                      Admin/Oper
----------- -------------------------------------- ------------
Bridge      fe80::f4f6:a4ff:fe97:3c88%Bridge/64    up/up
Ethernet0   2002::1/64                             up/down
dummy       fe80::90cd:eff:fe7b:359f%dummy/64      up/up
eth0        fe80::aa2b:b5ff:fe9d:d7db%eth0/64      up/up
lo          ::1/128                                up/up
```

Step 3. Save the setting to "config_db.json".

Use the **config save -y** command to save the configured IP addresses to the "**config_db.json**" file.

Example

```
admin@sonic:~$ sudo config save -y
```

Note

The modified **config_db.json** content will not be saved into the /etc/sonic/config_db.json file automatically. In order to do that, a **config save** command needs to be manually executed as shown in step 3.

Step 4. Check the IP address settings in "config_db.json".

Use the **vi /etc/sonic/config_db.json** command to check the IP address settings in the "**config_db.json**" file.

Example

```
admin@sonic:$ sudo vi /etc/sonic/config_db.json

 {
   ...omitted
   "INTERFACE": {
       "Ethernet0|192.168.1.1/24": {},
       "Ethernet0|2002::1/64": {},
   ...omitted
   },
   ...omitted
}
```

(202111.2) Enabling the Zero Touch Provisioning (ZTP) Service

Rev.02

Zero Touch Provisioning (ZTP)

Zero Touch Provisioning (ZTP) allows the automatic provisioning of new switches without any manual intervention. The ZTP service can be used by users to configure a fleet of switches using the common configuration templates.

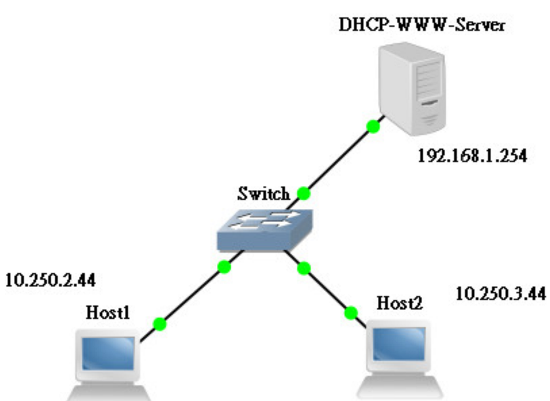
Switches booting from the factory default state will communicate with the remote provisioning server and perform the below set of actions.

- Obtains an IP address via DHCP
- Acquires a ZTP configuration file ZTP JSON[more] from a remote server that is specified in DHCP option 67 in IPv4 environments
- Executes the ZTP configuration (or plug-in) in the ZTP configuration file
- The switch configuration file[more] is downloaded if it's specified in the ZTP configuration or DHCP option.

This document describes how to enable and disable the ZTP service, or manually restart it. For detailed information about DHCPv6, DHCP options, other protocols, and provisioning scripts, please refer to [ztp.md in Community] and ZTP commands in the command reference.

Deployment Architecture

The below architecture diagram is referred to explain the configurations in the following sections.



In this architecture, a single server is used to provide the DHCP service and Web service for the HTTP protocol.

- The DHCP service provides an initial IP address for the switch and an URL for the ZTP JSON file.
- The Web service provides the ZTP JSON, scripts, and SONiC image.

After provisioning, the ZTP process may check connectivity by pinging Host1 and Host2. These hosts are shown in the above diagram.

The ZTP Service

The ZTP service is defined as a **systemd** service, which runs on the native SONiC O/S. This service is exited if the ZTP admin state is not "enable" and does not execute again until the next boot-up or if manually re-started. The ZTP service will not block other services, whether it is started or exited.

The following procedures are performed in the ZTP process.

1. The ZTP service will try to obtain management connectivity. It performs DHCP discovery on all connected in-band interfaces and the out-of-band management interface. The first received DHCP offer is used as a primary management interface to obtain user-provided provisioning data.
2. The switch will configure the IP address by DHCP and get a URL for the ZTP JSON file from DHCP Option 67.
3. The ZTP JSON is downloaded from the provisioning server to local. Each section of the ZTP JSON data is processed by the corresponding handler (the plugin).
4. The whole ZTP result (SUCCESS/FAILED) is determined by the result of each section.
5. The switch configuration file[more] should be present at the end.

Preparation

The following files should be prepared.

Provisioning server:

- ZTP JSON file: *ztp_data.json*. A local copy of the ZTP JSON file is maintained as the file */host/ztp/ztp_data.json*.
- The switch configuration file: *config_db.json*. Would be saved as */etc/sonic/config_db.json* on the switch.
- The SONiC image files: *sonic-xxx.bin*
- Provisioning script to do more complex configuration steps. (Optional)

DHCP server:

- DHCP server configuration file: *dhcpd.conf* [more] for ISC DHCP server.

Enable the ZTP Service

- When the ZTP service starts, it first checks if there already exists a ZTP JSON file locally, and performs the below set of actions.
  - if found, loads it.
  - If the *ztp.status* field of the local file is either SUCCESS, FAILED, or DISABLED, the ZTP service exits.
  - If the *ztp.status* field of the local ZTP JSON file is IN-PROGRESS or BOOT, the local file is used for further processing.
  - If no local ZTP JSON file is found, the ZTP service downloads the ZTP JSON file using the URL provided by DHCP Option 67 and starts processing it.
- There may be multiple sections in the ZTP JSON file as in the example[more]. They are handled by predefined or user-defined plugins[->]. The leading numbers '01-' or '02-' will decide the execution sequence of plugins.

Disable the ZTP Service

You can use the **config ztp disable** command to stop the ZTP service at any time and change the administrative mode to disabled. If ZTP is administratively disabled, the ZTP service does not run even after a reboot or if the startup configuration file is not present. You will have to use the **config ztp enable** command to enable it administratively again.[more]

Restart the ZTP Service

Application scenario: After the first time you boot up, the ZTP service properly installs the configuration file or upgrades the image. In case of future maintenance (e.g. you may need to install new configuration files for a switch), you can restart the ZTP service by using the config ztp enable command to enable it administratively and the config ztp run command to run the process.

The process will back up the old switch configuration file by renaming it to `config_db.json.[time_stamp]`.

Examples

ZTP JSON Example

In this example, the following actions will be performed:

1. Update the firmware (skip reboot)
2. Download the specified config_db.json file based on the serial number
3. Reboot the switch
4. Check connectivity

ztp_data.json

```
{
    "ztp": {
        "01-firmware": {
            "install": {
                "url": "http://192.168.1.254/ztp/image/sonic-broadcom.bin",
                    "skip-reboot": true
            }
        },
        "02-configdb-json": {
            "dynamic-url": {
                "source": {
                    "prefix": "http://192.168.1.254/ztp/config/",
                    "identifier": "serial-number",
                    "suffix": "_config_db.json"
                },
                "destination": "/etc/sonic/config_db.json"
            },
            "save-config" : true,
            "reboot-on-success": true
        },
        "03-connectivity-check" : {
            "ping-hosts" : [ "10.250.2.44", "10.250.3.44"]
        },
        "config-fallback": false,
        "halt-on-failure": false,
        "ignore-result": false,
        "reboot-on-failure": false,
        "reboot-on-success": false,
        "restart-ztp-no-config": true,
        "restart-ztp-on-failure": false
    }
}
```

Pre-defined ZTP plugins include "01-firmware," "02-configdb-json," and "03-connectivity-check". The order in which they execute is determined by the leading "01," "02," and "03" integers.

- 01-firmware: Download the SONiC image and run **sonic_installer set_default** by default. Skip the reboot is specified here.
- 02-configdb-json: Download config-db.json for the switch. Here, the serial number of the switch is utilized. This serial number is contained in the switch's HTTP request header. In this example, the file: "http://192.168.1.254/ztp/config/583554X2002010_config_db.json" is downloaded. After downloading, **config reload** is executed by default. And **config save** is specified here. After this task is executed successfully, reboot the switch.
- 03-connectivity-check: Uses ping to check the connectivity.

Finally, we can use the **sudo show ztp status** command to check the status.

ZTP status

```
admin@sonic:~$ sudo show ztp status
ZTP Admin Mode : True
ZTP Service    : Inactive
ZTP Status     : SUCCESS
ZTP Source     : dhcp-opt67 (Ethernet0)
Runtime        : 26m 48s
Timestamp      : 2021-01-06 07:01:45 UTC

ZTP Service is not running

01-firmware: SUCCESS
02-configdb-json: SUCCESS
03-connectivity-check: SUCCESS
```

DHCP Configuration Example

Using the ISC DHCP server and its configuration file `dhcpd.conf` as in the following example and the command **dhcpd -d -f -cf dhcpd.conf** is executed on the server.

dhcpd.conf

```
#
# Sample configuration file for ISC dhcpd for Debian
#
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as
# configuration file instead of this file.
#
#

# option definitions common to all supported networks...
option domain-name "example.org";
default-lease-time 600;
max-lease-time 7200;

log-facility local7;

subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.10 192.168.1.100;
#option 67 = 0x43
  if exists dhcp-parameter-request-list {
    option dhcp-parameter-request-list = concat(option dhcp-parameter-request-list,42);
    option dhcp-parameter-request-list = concat(option dhcp-parameter-request-list,43);
  }
  #option 66
  option tftp-server-name "192.168.1.254";
  #option 67
  option bootfile-name "http://192.168.1.254/ztp/ztp_data.json";

  #option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;

}
```

Troubleshooting

1. ZTP configuration file issue

Symptom

There is "ZTP JSON load failed" in syslog.

Action

Please check ztp_data.json in the server. It could be some type of typo in the ZTP configuration file.

For example, we use **#** to comment out a link. In JSON format, there is no comment-out symbol. If the line(s) is not necessary, please remove it.

Mal-configuration file

```
Mar  6 01:00:05.324681 Leaf-2 INFO sonic-ztp[1667]: Downloading provisioning data from http://10.249.34.236
/ztp.json to /var/run/ztp/ztp_data_opt67.json
Mar  6 01:00:05.687517 Leaf-2 ERR sonic-ztp[1667]: Exception [ZTP JSON load failed] occured while processing
ZTP JSON file /var/run/ztp/ztp_data_opt67.json.
Mar  6 01:00:05.687912 Leaf-2 ERR sonic-ztp[1667]: ZTP JSON file /var/run/ztp/ztp_data_opt67.json processing
failed.
```

2. ZTP service displays "Restarting network discovery after link scan." continuously on the console

Symptom

If you are using in-band ZTP, all in-band interfaces maybe are all link-down and repeating the message "Restarting network discovery after link scan". (Only out-of-band interface eth0 is link up)

```
Mar 17 02:30:51.099884 sonic INFO sonic-ztp[13888]: System is ready to respond.
Mar 17 02:30:51.330495 sonic INFO sonic-ztp[13889]: Link up detected for interface eth0
Mar 17 02:30:51.330925 sonic INFO sonic-ztp[13889]: Restarting network discovery after link scan.
Mar 17 02:30:52.330925 sonic INFO sonic-ztp[13889]: Restarting network discovery after link scan.
Mar 17 02:30:53.330925 sonic INFO sonic-ztp[13889]: Restarting network discovery after link scan.
...
```

Action

Method1: Modify model's hwsku.json to bring port link up

Please check the interface connection mode in the model's **hwsku.json**.

For example, the "fec" mode should be the same at both ends of the connection. If one is "none" and the other is "rs", the mismatching makes links down and the ZTP service always performs network discovery, trying to connect the DHCP server on the in-band interface.

For example, assume the DUT is AS7816 and the DHCP server is located on Ethernet0. To resolve the issue, you may **modify** DUT: /usr/share/sonic/device/x86_64-accton_as7816_64x-r0/Accton-AS7816-64X/hwsku.json.

append

```
    "Ethernet0": {
        "default_brkout_mode": "1x100G[40G]",
        "fec": "rs"
    },
...
```

then **execute** ztp disable/enable/run to bring the port link up.

Method2: Use SONiC CLI to bring port link up

1. Make sure the port speeds of DUT and the connected device are the same. Use 'show interfaces breakout' and 'config interface breakout' command to modify.
2. Make sure the port FEC modes are the same. Use 'sudo portconfig -p EthernetX -f [none|rs|fc]' to modify.
3. Make sure the port is 'admin up'. Use 'config interface startup' command to modify.
4. use 'show interface status' command to check status

3. Management vrf and ZTP

Symptoms

- During ZTP DHCP discovering, and configuring mgmt vrf with a user-defined plugin (script), it doesn't enable immediately.
- Network unreachable issue if the management vrf is enabled by the configdb-json plugin.

Action

- In a sonic implementation, mgmt vrf is not activated when ZTP DHCP discovery is enabled. The solution is to add the following entries in config_db.json and use the configdb-json plugin. The plugin will re-configure interfaces to stop ZTP discovery after downloading. Then mgmt vrf is enabled.

part of config_db.json

```
        "MGMT_INTERFACE": {
                "eth0|192.169.1.10/24": {
                        "gwaddr": "192.169.1.254"
                }
        },
        "MGMT_VRF_CONFIG": {
                "vrf_global": {
                        "mgmtVrfEnabled": "true"
                }
        }
```

- After mgmt vrf is enabled, the network command should specify the vrf 'mgmt' for the eth0 interface, otherwise the default vrf is used and can cause a network unreachable issue. A general method is to use 'ip vrf exec mgmt CMD', which will execute the CMD in vrf 'mgmt'.

Reference

Source Type	Description	Link
ZTP CLI	the ZTP commands	(202111) ZTP
ZTP HLD	the ZTP concept document	https://github.com/Azure/SONiC/blob/master/doc/ztp/ztp.md

(202111.2) Backing Up and Restoring the Default Configuration

This section describes different backup procedures for restoring the default Edgecore SONiC configuration. The backup configuration file is helpful if your device's configuration file has been misconfigured. The backup file allows you to define a known working configuration to which you can roll back at any time. The procedures are as below.

Manually Backing Up and Restoring the Default Configuration

After installing Edgecore SONiC, you can manually back up the default **config_db.json** configuration file by using the following command.

Example

```
admin@sonic:~$ sudo cp /etc/sonic/config_db.json /etc/sonic/config_db.json.bk
```

Use the following command, to restore the backed-up default configuration file.

Example

```
admin@sonic:~$ sudo cp /etc/sonic/config_db.json.bk /etc/sonic/config_db.json
```

After restoring the backed-up default configuration file, either use the command **config reload** to reboot the switch, or power-cycle the switch.

Example

```
admin@sonic:~$sudo config reload -y
```

Using the sonic-cfggen Command

You can use the **sonic-cfggen** command to generate a default configuration file that is identical to the file generated after ONIE installation. (For more information, see (202111.2) Restoring the Configuration to Factory Defaults.)

Example

```
sudo sonic-cfggen -H -p /usr/share/sonic/device/$Platform/platform.json --preset t1 -k $HwSKU > ~/default.log
```

> Note
>
> You can obtain the values of "$Platform" and "$HwSKU" from the **show platform** or **show version** commands.

Information in "show platform"
admin@sonic:~$ show platform summary Platform: x86_64-accton_as7816_64x-r0 HwSKU: Accton-AS7816-64X ASIC: Broadcom

Step 1: Generate the default config_db.json file in the home directory.

Example

```
admin@sonic:~$ sudo sonic-cfggen -H -p /usr/share/sonic/device/x86_64-accton_as7816_64x-r0/platform.json --
preset t1 -k Accton-AS7816-64X > ~/default.log
```

Step 2: Move the "default.log" file to the correct location.

Example

```
admin@sonic:~$ sudo cp default.log /etc/sonic/config_db.json
```

Step 3: Reboot or power cycle the switch.

Example

```
admin@sonic:~$sudo config reload -y
```

Using the restore.sh Script

You can back up the default configuration by uploading and executing the script **restore.sh** in SONiC. The script generates the default configuration **config_db.json** file automatically. Reboot the switch for the configuration to take effect.

Step 1: Set the management port IP address.

Example

```
admin@sonic:~$ sudo ip address add 188.188.98.21/16 dev eth0
```

Step 2: Upload the script to SONiC.

Example

```
admin@sonic:~$ sudo scp root@188.188.36.36:/root/restore.sh ~/
```

Step 3: Change permission of the script.

Example

```
admin@sonic:~$ chmod +x restore.sh
```

Step 4: Run the script.

Example

```
admin@sonic:~$ ./restore.sh
Get HwSKU and Platform from the database
/etc/sonic/config_db.json is restored to default
```

Step 5: Reboot or power cycle the switch.

Example

```
admin@sonic:~$sudo config reload -y
```

> **Note**
>
> Make sure to execute the **config reload** command to make the new configurations take effect.

(202111.2) Restoring the Configuration to Factory Defaults

This section describes the procedures to restore the configuration to factory defaults, that is, the configuration when the SONiC image was published.

Since the Dynamic Port Breakout (DPB) was introduced in version 202006 and later, the referred port configuration has changed from **port_config.ini** to **platform.json**. Depending on the installed version, you need to use the right option to generate the default configuration file. By default, the DPB option is used.

Restoring the Configuration from Templates

Configuration templates come from minigraph, yaml, and json, depending on the feature. Since the port configuration is model-dependent, it is recommended to use **sonic-cfggen** to generate the default port configuration from the template.

Command syntax

```
# Use DPB feature to generate the configuration (202006 or later branching)
sudo sonic-cfggen -H -p /usr/share/sonic/device/$Platform/platform.json --preset t1 -k $HwSKU > ~/<temp.log>


# Use port_config.ini to generate the configuration (201911 or earlier branching)
sudo sonic-cfggen -H -p /usr/share/sonic/device/$Platform/$HwSKU/port_config.ini --preset t1 -k $HwSKU > ~
/<temp.log>
```

Note

- <temp.log> store generated configuration.
- You can obtain the values of "$Platform" and "$HwSKU" from the **show platform** or **show version** commands.

Information in "show platform"

```
admin@sonic:~$ show platform summary
Platform: x86_64-accton_as7816_64x-r0
HwSKU: Accton-AS7816-64X
ASIC: broadcom
```

Example for Generating the Default Configuration

For versions 202006 and later

Example (using DPB feature)

```
admin@sonic:~$ sudo sonic-cfggen -H -p /usr/share/sonic/device/x86_64-accton_as7816_64x-r0/platform.json --
preset t1 -k Accton-AS7816-64X > ~/default.log
admin@sonic:~$ sudo cp default.log /etc/sonic/config_db.json
```

For versions before 202006

Example (using port_config.ini)

```
admin@sonic:~$ sudo sonic-cfggen -H -p /usr/share/sonic/device/x86_64-accton_as7816_64x-r0/Accton-AS7816-64X
/port_config.ini --preset t1 -k Accton-AS7816-64X > ~/default.log
admin@sonic:~$ sudo cp default.log /etc/sonic/config_db.json
```

Note

- Using **port_config.ini** to generate a configuration in 202006 (or later), the DPB function may not work as expected.
- Execute "sudo config reload -y" to make the new configuration take effect.

(202111.2) Contacting Technical Support

If you need any assistance with Edgecore SONiC, go to **support.edge-core.com** and submit a support ticket.

To help Edgecore's support process, be sure to include sufficient information on your SONiC installation. In particular, it helps to include one of the following:

- Show Techsupport
- Syslog
- Remote Syslog

Show Techsupport

The SONiC **show techsupport** command gathers information about a device and creates an archive file in `/var/dump/<DEVICE_HOST_NAME>_YYYYMMDD_HHMMSS.tar.gz`. This archive file can then be sent to Edgecore support to assist in resolving issues.

The **show techsupport** command can create a large file when a system has been running for a long time. To reduce the size of the file, the command can include a time limit option, as shown in the following example.

Example

```
show techsupport
admin@sonic:~$ show techsupport --since=yesterday  # Will collect syslog and core files for the last 24 hours
admin@sonic:~$ show techsupport --since='hour ago' # Will collect syslog and core files for the last one hour
```

Syslog

The **show logging** command displays all the currently stored log messages. All the latest processes and corresponding transactions are stored in the "syslog" file. This file is saved in the path `/var/log` and can be viewed using the **sudo cat syslog** command, which requires root login.

Example

```
admin@sonic:~$ show logging
```

Optionally, you can follow the log live as entries are written to it by specifying the -f (or follow) flag

Example

```
admin@sonic:~$ show logging --follow
```

Remote Syslog

There is no specific SONiC command to configure a remote syslog server. So, you must edit the `/etc/sonic/config_db.json` file to configure it. Follow the below steps for the configuration.

Step 1. Add the syslog server and its IP address to the **config_db.json** file.

Example

```
admin@sonic:/etc/sonic$ sudo vi config_db.json
{
    ...
    "SYSLOG_SERVER": {
        "192.168.1.3": {}
    },
    ...
}
```

Step 2. Reload the configuration or power cycle the switch.

Example

```
admin@sonic:~$ sudo config reload -y
```

Caution

Make sure the switch management port IP address has been configured to reach the server and connectivity has been verified. (Refer to (202111.2) Configuring IPv4/IPv6 Addresses for Ports.)

Syslog Severity Levels

By default, the switch sends all syslog messages (all severity levels) to the syslog server. The syslog severity levels are defined in the following table.

Value	Severity	Keyword
0	Emergency	emerg
1	Alert	alert
2	Critical	crit
3	Error	err
4	Warning	warning
5	Notice	notice
6	Informational	info
7	Debug	debug

To restrict messages sent to the syslog server by severity level, follow the below steps:

Step 1. Edit the file `/usr/share/sonic/templates/rsyslog.conf.j2`.

Default setting: all severity levels.

Example

```
admin@sonic:~$ sudo vi /usr/share/sonic/templates/rsyslog.conf.j2
...

#Set remote syslog server
{% for server in SYSLOG_SERVER %}
*.* @{{ server }}:514;SONiCFileFormat
{% endfor %}

...
```

Warning and below (Level 0 ~ 4).

Example

```
admin@sonic:~$ sudo vi /usr/share/sonic/templates/rsyslog.conf.j2
{ % for server in SYSLOG_SERVER % }
*.warning @{{ server }}:514;SONiCFileFormat
{ % endfor % }
```

Only warning level (Only Level 4).

Example

```
admin@sonic:~$ sudo vi /usr/share/sonic/templates/rsyslog.conf.j2
{% for server in SYSLOG_SERVER %}
*.=warning @{{ server }}:514;SONiCFileFormat
{% endfor %}
```

All severity levels except warning (Level 0 ~ 7 except Level 4).

Example

```
admin@sonic:~$ sudo vi /usr/share/sonic/templates/rsyslog.conf.j2
{% for server in SYSLOG_SERVER %}
*.debug;*.!=warning @{{ server }}:514;SONiCFileFormat
{% endfor %}
```

Step 2: Restart the syslog service.

Example

```
admin@sonic:~$ sudo systemctl restart rsyslog-config
```